Manual

Archiving of FEP Data

FEP-ARC 8.1

Version 1.0.1374

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Contents

1 Archiving of FEP Data

Purpose

The function package Archiving inspection data from the FEP makes it possible to access archived data and, moreover, to export collected data and to load these data back into the system as needed.

You use the function package when:

- You would like to have access to data already moved to archive tables in the function package.
- Because of legal requirements or customer demands, you need to ensure that the data entered is stored long-term.
- Because of legal requirements or customer demands, you need to import data back into the system to be evaluated / analyzed again.

Integration

The archiving of inspection requirements, incl. all referenced inspection data, is related to the components

- Inspection planning for in-process inspections
- Evaluations of in-process inspections
- Standard control cards and histograms
- Extended control cards and histograms, and
- Failure mode analysis / tracking of measures.

Features

- The following data areas / objects are archived
  - Inspection requirements
  - Inspection steps
  - Inspection points
  - Inspection data, e.g. measured values
  - Assigned errors
- Export function
  - Function for transferring (exporting) data from the archive tables to external file systems for the purpose of long-term storage of the recorded consumption data.
- Import function
  - Functions for importing exported data into the archive tables in order to evaluate them using the applications listed above.

2 Data Management

Overview

Menu	System administration → Archiving → Data management
Transaction code	arccfg
Function authorization	arccfg.*

Usage

You may use this application in order to view or edit the centrally managed settings for archiving.

Integration

The settings for archiving/for data management are made centrally from all components/functions in the application.

Field descriptions

Product

In the Product field, the HYDRA module for which a rule is to be defined is entered.

Object

The Object defines what is to be provided .

Retention period

In the "retention period" fields, you define how long the data should be available before being archived.

Unit

The unit for the retention period is indicated in days, months or years.

Last retention date

This field is computed and subsequently indicates the last day on which data are still available in this object.

Action

It is possible to select from three process variants for an object:

D = delete	Object is deleted
M = move (archive)	Object will be transferred to the next division
X = Export	Object is unloaded (XUNLOAD format), subsequently deleted

Target object

At present, this automatically results from the detail configuration.

Condition

The characters in this field are added to the database command controlling the action, whereby this condition is linked to the conditions of the original command there and is set in parenthesis.

Last run (date/time)

Indicates when this rule was applied last.

License

Indicates which license is required for archiving. Multiple licenses may be entered here (separated by a space). If none of these licenses is licensed on the system, the data are either deleted (action D) or unloaded in files (action M or X) according to the relevant action.

Path

Optional path for generating the file export (Unload). At present, only local drives are supported on the HYDRA server in the archive. If no path is set, file exports are filed as follows:

<HYDRADIR>/<SYSTEM>/custom/archive/<YYYY-MM-DD>/<PRODUCT>/

HYDRADIR … HYDRA directory

SYSTEM … System number

YYYY-MM-DD … Archiving date (YYYY … Year, MM … Month, DD … Day)

PRODUCT … Product from archiving configuration

Management table

Table name of management table where the archiving logs are stored.

Retention period management table

In the fields used for the Retention period, you define how long the logs should be available in the management table before being deleted.

Unit

The unit for the retention period is indicated in days, months or years.

Archiving step

Indicates whether this archiving executes archiving function I or II. In archiving function I, the data are transferred from the online inventory to the archive tables and/or deleted. Setting M (medium-term archive) In archiving function II, the data are transferred from the archive table to the file export and/or deleted. Setting L (long-term archive).

Configuration

Control indicator whether or not the configuration is active. Possible values: Y/N.

Archiving type

Identifier for time or object-related archiving. Supported modes: O = Object-related archiving (i.e. data are archived for each object individually). Z = Time-related archiving (i.e. data are collected/archived without any object reference). Please note: Time and object-related archiving differ from each other significantly in the archiving performance (runtime). Object-related archiving of mass data is not recommendable.

Priority

Integral value greater than 0. Indicates the execution sequence when several objects are defined to a module. Execution starts from the configuration with the lowest value.

Master table

Table including the archiving data. The extensions entered in the Condition field refer to this table.

Date column

Date column in the master table. Is used for evaluating the data to be archived with regard to the retention period.

Key 1

Clear key column in the master table; may be used to identify the data to be archived. Up to 5 key columns may be defined for object-related archiving. Time-related archiving only supports one primary key in Key 1.

Keys 2 – 5

Additional optional key columns for object-related archiving.

Comment

Comment line for describing the archiving configuration.

When copying the archiving configuration, only the data management configuration is copied at present, but no any defined data records from the object details. At present, the data records from the object details must be copied to the new configuration by the user in the appropriate screen.

Module	**Version**	**Description of archiving**
BDE	8.1	here
CAQ	8.1	here
MDE	8.1	here
MPL	8.1	here

3 CAQ Specific Configurations

3.1 Overview

3.1.1 Archiving inspection requirements

3.1.1.1 Data structures

By default, inspection requirements are archived by object, which means that each individual inspection requirement is taken into account during archiving. If it meets the conditions of the configured parameters, then it will be archived. During this process, all of the corresponding detail data are archived as well.

Optionally, the same mechanism can be used to configure that the data are deleted (instead of archiving the data).

In the standard version, you will find a separate configuration for each individual HYDRA-CAQ data type (in-production inspection, goods receipt or goods issue inspection, initial sample inspection, QMS data). This allows you, for example, to archive the inspection requirements from the goods receipt in other intervals than data from the production area.
However, by default, the details backed up for the inspection requirement are the same for every data type. In detail, this includes the following data:

Data	Source table(s)	
Inspection requirements	caq_pruefanf caq_pan_zusatz	
Inspection orders	caq_paukop	
Inspection order configurations Characteristics configurations	caq_paukonf	
Inspection points	caq_numpool caq_ppktm_info	
QMS: Dynamic modification history for inspection points based on characteristics	caq_dyhis_ppktmm	
Characteristics	caq_merkmal caq_merk_zusatz	
Inspection frequencies	caq_prueffreq	
Quantity-dependent inspection specifications	caq_mengabh_prf	
Documents	caq_dokus	
Tool assignments	caq_werkzzuord	
Samples Results of characteristics	caq_paustich	
Sample numbers assignment Samples inspection point assignment	caq_paunumm	

Data	Source table(s)	
Characteristic attributes	caq_paumm_ausp	
Single values	caq_paumwert	
Failure analysis entries	caq_fhlanal	
Measures and corresponding parameters	caq_massn caq_mass_param	
Inspection matrix Characteristic inspection point assignments	caq_pruefmatrix	

Events and logging entries belonging to the inspection requirements are archived separately. The archiving configuration for this data is described in the following chapters.

By default, the inspection requirements for the PMV data types (calibration/ maintenance) are not archived.

3.1.1.2 Standard configuration (with license FEP/WEP/WAP/QMS-ARC)

If the one of the licenses FEP/WEP/WAP/QMS-ARC is active for the respective data type, then, by default the inspection requirements for the data types described below will be backed up in accordance with the previously described structure.
There is a separate configuration for each of these data types that can be used to set the parameters for separate archiving periods, for example.

The standard configuration for the inspection requirements is designed for two-step archiving.

In the first step, data is moved into the medium-term data area. After that, they are directly available for reports from the medium-term data area. However, no more changes can be made to the data.
In the second step, data is moved into the long-term data area. After that, they are no longer directly available for reports. To make the data available for HYDRA reports, they must first be uploaded again.
The default configurations used for the first and second steps of archiving inspection requirements and their corresponding intervals are described below.

Product	Object	Description of the action	Factory default interval
CAQ	FEP	Moving the production inspection requirements from the online database to the medium-term database	1 year
CAQ	A_FEP	Moving the production inspection requirements from the medium-term database to the long-term database	3 years
CAQ	WEP	Moving the goods receipt inspection requirements from	1 year

Product	Object	Description of the action	Factory default interval
		the online database to the medium-term database	
CAQ	A_WEP	Moving the goods receipt inspection requirements from the medium-term database to the long-term database	3 years
CAQ	WAP	Moving the goods issue inspection requirements from the online database to the medium-term database	1 year
CAQ	A_WAP	Moving the goods issue inspection requirements from the medium-term database to the long-term database	3 years
CAQ	EMU	Moving the initial sample inspection requirements from the online database to the medium-term database	1 year
CAQ	A_EMU	Moving the initial sample inspection requirements from the medium-term database to the long-term database	3 years
QMS	QMS	Moving the QMS inspection requirements from the online database to the medium-term database	3 months
QMS	A_QMS	Moving the QMS inspection requirements from the medium-term database to the long-term database	3 years

Used as the time reference field to calculate the intervals is the editing date for the inspection requirement. However, not all inspection requirements are archived. Whether or not an inspection requirement is archived depends on its status. By default, only those inspection requirements are archived that are finished and canceled.

In addition, QMS group requests must be uploaded to the PPS system before they can be archived.

In the standard configuration, administration data relating to archived CAQ inspection requirements are maintained in the arc_verw_caq table for 12 years.

3.1.1.3 Actions without the FEP/WEP/WAP/QMS-ARC license

If there is no active FEP/WEP/WAP/QMS-ARC license, the inspection requirements are not affected by this archiving configuration.

As opposed to other HYDRA data, in this case, the inspection requirements are not removed. As such, they remain in the online data area permanently, unless they are accounted for in a different archiving configuration.

The inspection requirements for the QMS data are the exception. Because the events for these inspection requirements are usually uploaded to the PPS system, there is no urgent need to leave the data redundant in HYDRA.

This is why the data, as is the case in other HYDRA data areas, are removed from the system after the period of time described in the Chapter Default configuration (with license FEP/WEP/WAP/QMS-ARC) for the QMS product, object QMS, unless the configuration was changed otherwise.

3.1.2 Archiving group requests

3.1.2.1 Data structures

By default, group requests are archived by object, which means that each individual group request is taken into account during archiving. If it meets the conditions of the configured parameters, then it will be archived. During this process, all of the corresponding detail data are archived as well.

Optionally, the same mechanism can be used to configure that the data are deleted (instead of archiving the data).

As opposed to how inspection requirements are archived, group requests are not archived by data type. Therefore there is also no way to parameterize different intervals for group requests of different data types.

By default, the backed up details for a group request contain the following data:

Data	Source table(s)
Group requests	caq_sammelanf
Slow user fields	caq_zusatz_feld
Inspection frequencies	caq_prueffreq

Logging entries belonging to the group requests are archived separately. The archiving configuration for this data is described in the following chapters.

3.1.2.2 Standard configuration (for archiving with CAQ license)

If the archiving license is active for the respective data type, then by default the group requests, depending on their data type, will be backed up in accordance with the previously described structure.

The standard configuration for the group requests is designed for two-step archiving.

In the first step, data is moved into the medium-term data area. After that, they are directly available for reports from the medium-term data area. However, no more changes can be made to the data.
In the second step, data is moved into the long-term data area. After that, they are no longer directly available for reports. To make the data available for HYDRA reports, they must first be uploaded again.
The default configurations used for the first and second steps of archiving group requests are described below.

Product	Object	Description of the action	Factory default interval
CAQ	SAN	Moving the group requests from the online database to	1 year

Product	Object	Description of the action	Factory default interval
		the medium-term database	
CAQ	A_SAN	Moving the group requests from the medium-term database to the long-term database	3 years

Used as the time reference field to calculate the intervals is the editing date for the group request.

By default, only those group requests are archived in the medium-term data area for which there are no more inspection requirements in the online area. This ensures that a group request is not archived in the medium-term data area until all of the corresponding inspection requirements have also been archived.

Archiving in the long-term data area follows the same principle. The group request (accounting for the interval) is not moved into the long-term data area until there are no more inspection requirements belonging to a group request in the medium-term data area.

In the standard configuration, administration data relating to archived CAQ group requests are maintained in the arc_verw_caq table for 12 years.

3.1.2.3 Actions for archiving without CAQ license

If there is no active CAQ license for the relevant data type, the group requests for this archiving configuration remain untouched.

As opposed to other HYDRA data, in this case, the group requests are not removed. As such, they remain in the online data area permanently, unless they are accounted for in a different archiving configuration.

3.1.3 Archiving CAQ events

3.1.3.1 Data structures

By default, CAQ events are archived sorted by time. Data is archived irrespective of whether or not the corresponding CAQ objects (e.g. inspection requirements or their details) were archived.

Optionally, the same mechanism can be used to configure that the data are deleted (instead of archiving the data).

By default, the events are archived individually, in some cases with corresponding detail data. In detail, this relates to the data described below:

Data	Source table(s)
CAQ events	event_caq
Optional dialog data corresponding to the events	event_dlg_data

3.1.3.2 Standard configuration (for archiving with CAQ license)

If the archiving license is active for the respective data type, then by default the CAQ events, depending on their data type, will be backed up in accordance with the previously described structure.

The standard configuration for the CAW events is designed for two-step archiving.

In the first step, data is moved into the medium-term data area. After that, they are directly available for reports from the medium-term data area.

In the second step, data is moved into the long-term data area. After that, they are no longer directly available for reports. To make the data available for HYDRA reports, they must first be uploaded again.

The default configurations used for the first and second steps of archiving group requests are described below.

Data	Source table(s)
CAQ events	event_caq
Optional dialog data corresponding to the events	event_dlg_data

Used as the time reference field to calculate the intervals is the date of the CAQ events.

In the standard configuration, administration data relating to archived CAQ events are maintained in the arc_verw_caq table for 12 years.

3.1.3.3 Actions for archiving without CAQ license

If there is no active CAQ license for the relevant data type, the CAQ events for this archiving configuration remain untouched.

As opposed to other HYDRA data, in this case, the CAQ events are not removed. As such, they remain in the online data area permanently, unless they are accounted for in a different archiving configuration.

3.1.4 Archiving CAQ logging entries

3.1.4.1 Data structures

By default, CAQ logging entries are archived sorted by time. Data is archived irrespective of whether or not the corresponding CAQ objects (e.g. inspection requirements or their details) were archived.

Optionally, the same mechanism can be used to configure that the data are deleted (instead of archiving the data).

By default, the logging entries are archived individually, in some cases with corresponding detail data. In detail, this relates to the data described below:

Data	Source table(s)
Logging entries	hyd_logging
Additional data corresponding to logging entries	hyd_logging_data

3.1.4.2 Standard configuration (for archiving with CAQ license)

If the archiving license is active for the respective data type, then, by default the CAQ logging entries will be backed up in accordance with the previously described structure.

The standard configuration for the CAQ logging entries is designed for two-step archiving.

In the first step, data is moved into the medium-term data area. After that, they are directly available for reports from the medium-term data area.

In the second step, data is moved into the long-term data area. After that, they are no longer directly available for reports. To make the data available for HYDRA reports, they must first be uploaded again.

The default configurations used for the first and second steps of archiving CAQ logging entries are described below.

Product	Object	Description of the action	Factory default interval
CAQ	LOG	Moving the CAQ logging entries from the online database to the medium-term database	35 days
CAQ	A_ LOG	Moving the CAQ logging entries from the medium-term database to the long-term database	3 years

Used as the time reference field to calculate the intervals is the editing date of the logging entry.

In the standard configuration, administration data relating to archived CAQ logging entries are maintained in the arc_verw_caq table for 12 years.

3.1.4.3 Actions for archiving without CAQ license

If there is no active CAQ license for the relevant data type, the CAQ logging entries for this archiving configuration remain untouched.

As opposed to other HYDRA data, the CAQ logging entries are not removed in this case. As such, they remain in the online data area permanently, unless they are accounted for in a different archiving configuration.

4 Reload Manager

Summary

Menu	System Administration → Archiving → Reload Manager
Transaction code	arcrld
Function authorization	arcrld arcrld.export arcrcl.import

Utilization

The customer is responsible for backing up data in a data archive and for restoring data. When data is backed up it is transferred from the long-term data area to the archive data area and, as a result, it is stored in a separate file system. The customer can start the backup process using the Reload Manager.

When data, which is filed in the archive data area, is restored, it is transferred back (copied) to the long-term data area. The customer has to do this and bears the responsibility.

To be able to evalaute restored data using standard evaluations/reports, the data has to be transferred back (copied) to the reload data area.

For this reason, the Reload Manager enables the following functions:

1. Moving of exported data into the customer archive (the customer archive path has to be defined within the HYDRA path settings).
2. Loading of exported data into the reload area for evaluations

Integration

The Reload Manager is a central function that is used by many components or functions.

Selection Criteria

The application provides the following selection criteria:

Module

Reference to the product group which archived data belongs to.

Object type

Object type of the archived data.

The selection options “module” and “object type” are mandatory fields.

Toolbar

Export

The “EXPORT” button allows for the entries selected in the Reload Manager to be exported to the customer directory (all data records can be selected using the context menu of the right mouse button). Once the function has been started, the input dialog that opens requires the customer archive path to be entered.

Import

The “IMPORT” button allows for the selected entries to be loaded into the reload area, in order for the data to be again available for HYDRA evaluations/reports. Once the function has been started, the data loading mode and an optional path are requested. By indicating the path, it is possible to specify another storage location for the files, provided that the archived files have not been moved to the customer-specific archive using the Reload Manager.

There are three different ways to deal with reload data after they have been reloaded to prevent the reload data set from increasing excessively (→ no slow evaluations/reports):

The following modes are distinguished:

- Cyclic
  Data is loaded to the reload area in the “cyclic” mode. Loaded data is automatically removed from the corresponding reload tables, once the retention period specified within archive settings has expired.
  Please compare the configuration entry HYD / RELOADMANAGER of the data management function. When demo settings are used, data is removed from the reload area after 14 days.

  When data is imported, the HYD_REL_MANAGEMENT.DELETE_DATE field is calculated subject to the configuration and entered in the reload management table.

- User-specific
  Data is loaded to the reload area in the “user-specific” mode. The loaded data is automatically removed from the corresponding reload area, once the time specified for the user within the user-specific settings has expired. Exception: In case several users loaded identical data into the reload area, the corresponding retention period is automatically set to the maximum date.
  When data is imported, the HYD_REL_MANAGEMENT.DELETE_DATE field is computed subject to the configuration and entered in the reload management table.

- Manual

  Data is loaded in the “manual” mode to the reload area. The customer is responsible for deleting data from corresponding reload tables. However, identical data cannot be loaded several times in the “manual” mode (data can be loaded in the “user-specific” mode though).

In order for data to be transferred to the reload data area, the HYDRA server must be able to access these files. Otherwise, loading is cancelled with an error message.